

GATE Previous Year Questions

Topic: Stacks & Queues | Years: 2020–2024 | Total: 50 Questions

STACK — BASICS

Q1. [GATE 2024]

Which of the following applications uses a stack?

(A) BFS traversal (B) CPU Round Robin scheduling (C) Recursive function calls (D) Printer spooling

Q2. [GATE 2024]

A stack is used to convert infix expression $P + Q * R - S$ to postfix. What is the result?

Q3. [GATE 2024]

Evaluate the postfix expression: $4\ 5\ +\ 3\ * \ 2\ -$

Q4. [GATE 2023]

The Next Greater Element for every element in array $[4, 5, 2, 10]$ using a stack gives?

Q5. [GATE 2023]

The minimum number of stack operations to sort a stack in ascending order using one additional stack is of order?

Q6. [GATE 2023]

Consider a stack with operations: $\text{push}(1), \text{push}(2), \text{push}(3), \text{pop}(), \text{push}(4), \text{pop}(), \text{pop}()$. What is the output sequence?

Q7. [GATE 2023]

Which data structure efficiently checks if parentheses $\{ [()] \}$ are balanced?

Q8. [GATE 2022]

Evaluate the prefix expression: $- * 4\ 3\ +\ 2\ 1$

Q9. [GATE 2022]

A stack is implemented using a singly linked list. What is the time complexity of push and pop operations?

Q10. [GATE 2022]

Convert infix $(A - B / C) * (D + E * F)$ to postfix.

Q11. [GATE 2022]

What is the maximum size of the stack needed to evaluate the postfix expression $a\ b\ c\ d\ +\ * \ -\ ?$

Q12. [GATE 2021]

Which traversal of a binary tree uses a stack in iterative implementation?

Q13. [GATE 2021]

A function is called recursively 5 times before returning. How many activation records does the call stack hold at maximum depth?

Q14. [GATE 2021]

Tower of Hanoi for $n = 4$ disks requires minimum how many moves?

Q15. [GATE 2021]

Largest rectangle in a histogram of n bars is solved optimally using which data structure and what time complexity?

Q16. [GATE 2020]

In a stack of size 5, after push(10), push(20), push(30), pop(), push(40), what is the current top element?

Q17. [GATE 2020]

Find the Next Greater Element for every element in array [2, 7, 3, 5, 1].

Q18. [GATE 2020]

DFS on a graph uses which data structure?

STACK — SEQUENCES & PERMUTATIONS

Q19. [GATE 2024]

Elements 1, 2, 3, 4 are pushed in order. Is pop sequence 2, 4, 3, 1 valid? Justify.

Q20. [GATE 2024]

Elements 1, 2, 3, 4 are pushed in order. Is pop sequence 4, 1, 2, 3 valid? Justify.

Q21. [GATE 2023]

For $n = 5$ elements pushed in order, how many distinct valid pop sequences are possible?

Q22. [GATE 2022]

Push sequence 1, 2, 3. Which pop sequence is NOT possible?

(A) 1,2,3 (B) 2,1,3 (C) 3,2,1 (D) 3,1,2

Q23. [GATE 2021]

Push order: 1,2,3,4,5. Is pop sequence 3,2,1,4,5 valid? Justify.

Q24. [GATE 2020]

How many permutations of {1,2,3} are obtainable as pop sequences from a stack (push order 1,2,3)?

QUEUE — BASICS

Q25. [GATE 2024]

BFS of a graph requires which data structure?

Q26. [GATE 2024]

In a circular queue of size 6, front = 3, rear = 1. How many elements are present?

Q27. [GATE 2024]

A queue is used in which CPU scheduling algorithm?

Q28. [GATE 2023]

Level-order traversal of a binary tree uses which data structure?

Q29. [GATE 2023]

In a queue using an array of size N , what is the condition for EMPTY in a circular queue?

Q30. [GATE 2023]

What is the time complexity of enqueue and dequeue in a queue implemented via linked list?

Q31. [GATE 2022]

A printer spooler queuing print jobs is an example of which data structure?

Q32. [GATE 2022]

In a circular queue of size 7, after 5 enqueues and 2 dequeues, $\text{front} = 2$, $\text{rear} = 5$. Is the next enqueue possible?

Q33. [GATE 2022]

What is the maximum usable capacity of a circular queue of array size N ?

Q34. [GATE 2021]

Which application uses a queue?

- (A) Undo operation (B) Balancing parentheses (C) Handling keyboard buffer (D) Evaluating postfix expressions

Q35. [GATE 2021]

In BFS of graph starting from node A (adjacency: $A \rightarrow B, C$; $B \rightarrow D$; $C \rightarrow D$), what is the order of visiting nodes?

Q36. [GATE 2021]

Which data structure supports $O(1)$ insertion at rear and $O(1)$ deletion at front?

Q37. [GATE 2020]

A queue is implemented using two stacks S_1 and S_2 . Enqueue pushes onto S_1 ; dequeue pops from S_2 (transferring S_1 to S_2 if S_2 is empty). What is the amortized cost per operation?

Q38. [GATE 2020]

In OS, which queue holds processes waiting for I/O completion?

DEQUE & PRIORITY QUEUE

Q39. [GATE 2024]

A deque allows insertion and deletion at which ends?

Q40. [GATE 2024]

Find the sliding window maximum of array $[1, 3, -1, -3, 5, 3, 6, 7]$ with window size 3.

Q41. [GATE 2023]

Dijkstra's algorithm achieves $O((V+E) \log V)$ using which data structure?

Q42. [GATE 2023]

In a Max-Heap used as a priority queue, where is the maximum element always located?

Q43. [GATE 2022]

What is the time complexity of insertion into a binary min-heap of n elements?

Q44. [GATE 2022]

What is the time complexity of delete-min from a binary min-heap of n elements?

Q45. [GATE 2021]

What is the time complexity of building a heap from n unsorted elements using bottom-up heapification?

Q46. [GATE 2021]

A priority queue is used in which algorithm?

(A) DFS (B) BFS (C) Prim's MST (D) Topological Sort

Q47. [GATE 2020]

Which deque variant allows deletion only at front but insertion at both ends?

Q48. [GATE 2020]

Which deque variant allows insertion only at rear but deletion at both ends?

MIXED / APPLICATION QUESTIONS

Q49. [GATE 2023]

Processes P1, P2, P3 arrive in order with burst times 4, 2, 3 under FCFS scheduling (queue-based). What is the average waiting time?

Q50. [GATE 2022]

A stack must support push, pop, and get-minimum in $O(1)$ time. What data structure design achieves this?